

Solving the Shortest Vector Problem in $2^{0.63269n+o(n)}$ time on Random Lattices

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Abstract. The Shortest Vector problem (SVP) is one of the most important problem in lattice-based cryptanalysis. There is currently a gap in the understanding of this problem with respect to its worst-case complexity and its average-case behaviour. For instance, SVP has worst-case complexity $2^{n+o(n)}$ on an n -dimensional lattice [ADRS15]. However, in practice, people rely on heuristic (unproven) sieving algorithms of time complexity $2^{0.292n+o(n)}$ [BDGL16] to assess the security of lattice-based cryptography schemes. Those heuristic algorithms are experimentally verified for lattices used in cryptography, which are usually random in some way ^{*}.

In this paper, we try to bridge the gap between worst-case and heuristic algorithms. Using the formalism of random real lattices developed by Siegel [Sie45], we show a tighter upper bound on an important lattice parameter called the smoothing parameter that applies to almost all random lattices. Using a known discrete Gaussian sampler at the smoothing parameter, we can then directly sample short vectors. This allows us to provably solve an approximation version of the SVP on almost all random lattices with a small constant approximation factor 1.123, in time $2^{n/2+o(n)}$. With further analysis, we can also provably solve the exact SVP in time $2^{0.63269n+o(n)}$ on most random lattices. We also provide a smooth time/approximation factor tradeoff between these two cases. All our algorithms work in space $2^{n/2+o(n)}$.

1 Introduction

Lattice problems are central to modern cryptography and computational complexity theory due to their inherent hardness, which provides a foundation for secure cryptographic protocols. These problems are believed to be difficult to solve efficiently, even for quantum computers. This makes lattice-based cryptography a promising candidate for post-quantum security, offering resilience against future quantum attacks [BDK⁺18, DKL⁺18, FHK⁺19]. Moreover, lattice problems have applications in algorithmic number theory [LLL82], convex optimization [Jr.83, Kan87, FT87], coding theory [dB89], and cryptanalysis tools [Sha84, Bri84, LO85], reinforcing their importance across both theoretical and practical domains in computer science.

There is currently a gap in the understanding of these problems with respect to their worst-case complexity and their average-case behaviour. For instance, the Shortest Vector problem (SVP) on an n -dimensional lattice has worst-case complexity $2^{n+o(n)}$ [ADRS15]. However, in practice, people rely on heuristic (unproven) sieving algorithms of time complexity $2^{0.292n+o(n)}$ [BDGL16] to assess the security of lattice-based cryptography schemes. Those heuristic algorithms are experimentally verified for lattices used in cryptography, which are usually random in some way, but only seem to provide very short and not shortest nonzero vectors.

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^{*}There exists several formal notions of random lattices.



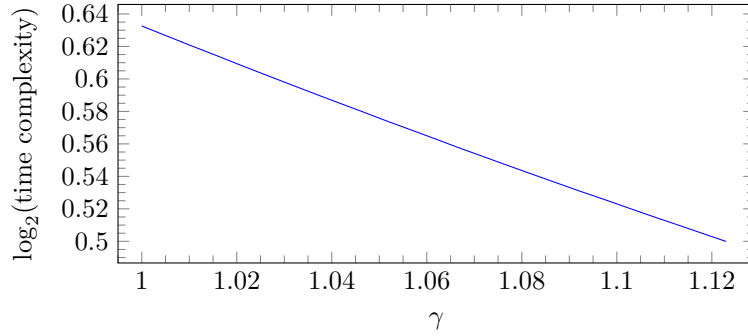


Figure 1: Time complexity to solve γ -SVP on random lattices (Theorem 1)

For most cryptographic applications, finding a short, but not necessarily shortest, nonzero vector is in fact sufficient. The γ -SVP consists in finding a nonzero vector of length at most γ times the length of a shortest nonzero vector. Surprisingly, very little is known about the worst-case complexity about this problem. The best provable algorithm has worst-case complexity $2^{0.802n+o(n)}$ [WLW15] but only solves $O(1)$ -SVP for an unspecified constant. More explicit constants are provided in [AUV19] but even for $\gamma = 100$, the time complexity is still $2^{0.824n+o(n)}$. In [BN24], the authors noted that there is no theoretical evidence to show that $\sqrt{2}$ -SVP is easier than SVP. In fact, solving $\sqrt{2}$ -SVP in time better than $2^{n+o(n)}$ would give a better algorithm[†] for ZLIP which is a recent hardness assumption in lattice-based cryptographic schemes [DvW22, BGPSD23]. If we relax the approximation ratio, better complexities can be achieved. For example, [ALSD21] gives an algorithm that solves $\tilde{O}(\sqrt{n})$ -SVP in time $2^{n/2+o(n)}$.

SVP In this paper, we try to bridge the gap between worst-case and heuristic algorithms. Using the formalism of random real lattices developed by Siegel [Sie45], we obtain a $2^{n/2+o(n)}$ time algorithm for 1.123-SVP and a $2^{0.63269n+o(n)}$ time algorithm for exact SVP on random lattices. More generally, we obtain a smooth tradeoff between the time complexity and the approximation factor (see Figure 1). Our algorithm (Theorem 1) achieves a much better approximation ratio compared to all worst-case algorithms and is particularly simple compared to [ALSD21]. For technical reasons, we had to separate the result on SVP (Theorem 11) from the one on γ -SVP (Theorem 10), due to the analysis being different for SVP.

Theorem 1 (Informal, see Corollary 6, Corollary 8 and Remark 4). *There is an algorithm that, for every $n \geq 1$ and $\gamma \in [1, 1.122973948]$ solves γ -SVP on most lattices in time $e^{o(n)}(\frac{\gamma^2}{2}e^{-\gamma^2/2e})^{-n/2}$ and space $2^{n/2+o(n)}$.*

Hermite SVP Most lattice reduction algorithms rely on solving γ -SVP in smaller dimension [Sch87] for γ close to 1 [ABLR21]. In certain lattice reduction algorithms such as slide reduction [GN08, ALNSD20], it is more convenient for the analysis to compare the length of short vectors to the determinant of the lattice instead of the length of a shortest vector. The problem of finding a vector of length at most $\alpha \text{vol}(\mathcal{L})^{1/n}$ for a lattice $\mathcal{L} \subset \mathbb{R}^n$ is known as the α -Hermite SVP (α -HSVP). Our algorithm (Theorem 2) also solves $\sqrt{\frac{0.2320n}{\pi}}$ -HSVP in time $2^{n/2+o(n)}$ and $\sqrt{\frac{n}{2\pi e}}$ -HSVP in time $2^{0.63269n+o(n)}$ for random

[†]Indeed, [BN24] showed that is a reduction from ZLIP in dimension n to $\sqrt{2}$ -SVP in dimension $n/2$, which together with [ADRS15] implies a $2^{n/2+o(n)}$ algorithm for ZLIP, the best currently known.

lattices. More generally, we obtain a smooth trade-off between the time complexity and the approximation factor. This improves upon the worst-case algorithm of [ALSD21] which solves $\tilde{O}(\sqrt{n})$ -HSVP in time $2^{n/2+o(n)}$. Indeed, we avoid the extra logarithmic factors in the approximation ratio and obtain a much better constant.

Theorem 2 (Informal, see Theorem 9 and Corollary 4). *There is an algorithm that, for every $n \geq 1$ and $\beta \in [\frac{1}{2e}, 0.2320]$ solves $\sqrt{\frac{n\beta}{\pi}}$ -HSVP on most lattices in time $e^{o(n)}(\beta e^{1-\beta})^{-n/2}$ and space $2^{n/2+o(n)}$.*

All our algorithms work by sampling the discrete Gaussian distribution $D_{\mathcal{L},s}$ on the lattice $\mathcal{L}$, a very commonly used distribution in lattice algorithms. This distribution is parametrized by the width s . It is known that sampling from $D_{\mathcal{L},s}$ is easy when s is large but very hard when s is small. An important quantity, known as the smoothing parameter $\eta_\varepsilon(\mathcal{L})$, intuitively characterizes when $D_{\mathcal{L},s}$ transitions from a “smooth” distribution to a discrete one (see Section 3.5). In particular, we use [ADRS15] to sample from $D_{\mathcal{L},s}$ for $s = \sqrt{2}\eta_{1/2}(\mathcal{L})$. An important part of the proof is estimating the quantity $\eta_{1/2}(\mathcal{L})$ which is a hard problem in general. This is where we can take advantage of random lattices by obtaining a tighter bound than for general lattices (see below).

The sampler from [ADRS15] takes time $2^{n/2+o(n)}$ but produces $2^{n/2+o(n)}$ vectors. Interestingly, we find that already with a single call to the sampler we can solve 1.123-SVP. By calling the sampler an exponential number of times, we can decrease the approximation factor and even solve SVP. Therefore, any improvement in the complexity of those samplers would yield an improvement to the complexity of solving 1.123-SVP (and correspondingly HSVP) for random lattices. Lattice-based cryptography relies on the fact that problems such as γ -SVP for small γ are hard, even for random lattices, with no subexponential-time algorithms. On the other hand, sampling efficiently from the discrete Gaussian distribution at the smoothing parameter is still an open problem, with no subexponential algorithm currently known. Therefore, we can view our result as an average-case hardness result for discrete Gaussian sampling (DGS) at the smoothing parameter. Plainly, solving DGS at the smoothing parameter in subexponential time for random lattices would have a major impact in lattice-based cryptography.

Probabilistic bound on the smoothing parameter Our main technical result is a probabilistic bound on the value of $\eta_\varepsilon(\mathcal{L})$ for a random lattice $\mathcal{L}$ for all $\varepsilon > 0$. This allows us to obtain tighter bounds on the length of vectors sampled from $D_{\mathcal{L},s}$, compared to the case of general lattices. More precisely, we obtain probabilistic bounds on the Gaussian mass $\rho_s(\mathcal{L})$ for a random lattice, and more generally on $\rho_s(\mathcal{L} \cap B_n(r))$ for any $r > 0$ (see Section 3.5). The latter could be of independent interest since tail bounds on the Gaussian mass are extremely useful in the analysis of lattice algorithms.

Formalisation of the Gaussian heuristic (GH) The GH states that the length of a shortest vector of a random lattices is approximately $\sqrt{\frac{n}{2\pi e}} \text{vol}(\mathcal{L})^{1/n}$. This heuristic has been shown to be provably true for various notions of random lattices (see Section 4.3). In this paper, we improve upon the state of art by obtaining bounds on λ_1 with explicit constants (Theorem 8) for random real lattices.

Approximate GapSVP The security of certain cryptographic primitives, such as fully homomorphic encryption [BV14], can be based on the *worst-case* hardness of a decision version of the approximate SVP problem, known as γ -GapSVP. This problem asks to decide, for a given $r > 0$, if a lattice $\mathcal{L}$ satisfies $\lambda_1(\mathcal{L}) \leq r$ or $\lambda_1(\mathcal{L}) \geq \gamma r$ (note that this is a promise problem). The hardness of this problem has been studied extensively as well and the best known result is a $2^{n/2+o(n)}$ -time algorithm for 1.93-GapSVP [ADRS15].

In this paper, we study *random lattices*, therefore it is natural to wonder if γ -GapSVP remains hard in this setting. This should intuitively not be the case because by the Gaussian Heuristic, the value of λ_1 for a random lattice is expected to be close to $\sqrt{\frac{n}{2\pi e}} \text{vol}(\mathcal{L})^{1/n}$, which makes the problem easy to decide. We formally prove that this is the case by giving a polynomial-time algorithm (Theorem 3) that solves γ -GapSVP on most lattices, for any $\gamma > 1$. Although this is more of a folklore result, our theorem gives explicit constants and bounds not found in the literature.

Theorem 3 (Informal, see Theorem 12 and Corollary 9). *There is an algorithm that, for every $\gamma > 1$, $n \geq 1$ and on most lattices $\mathcal{L} \subset \mathbb{R}^n$, solves γ -GapSVP in polynomial time.*

Bound on the variance of lattice sums for random lattices Most results on real random lattices are derived from the fundamental work of Siegel (see Section 4). This results allow to compute the expected value of sums of the form $\sum_{x \in \mathcal{L}} f(x)$ for a random lattice $\mathcal{L}$. However, in certain cases, the expected value is insufficient to conclude and higher moments are necessary. Rogers' theorem generalizes Siegel averaging theorem to more general sums of the form $\sum_{x_1, \dots, x_n \in \mathcal{L}} f(x_1, \dots, x_n)$. Unfortunately, the expressions quickly become difficult to work with, even for the simple case of trying to compute the variance of $\sum_{x \in \mathcal{L}} f(x)$. In this paper, we obtain a corollary of Rogers' theorem precisely for the case of the variance of a simple sum. The expression is significantly simpler to work with and appears to be new (Corollary 1).

Organization of the paper Section 2 contains some discussion and open problems. Section 3 contains preliminary technical results. Section 5 gives some probabilistic bounds on the Gaussian mass and smoothing parameter of random real lattice. Section 6 give an application of those bounds to the approximate (Hermite) SVP.

2 Discussion and open questions

Consequences for cryptography In this paper, we obtain a conceptually simple algorithm to solve (approximate) SVP and HSVP for random lattices by discrete Gaussian sampling. Perhaps the most intriguing consequence of these results is that it implies that sampling from a discrete Gaussian at the smoothing parameter cannot be done in subexponential time (even for random lattices) without major consequences on lattice-based cryptography. This, however, does not quite settle the question of the exact complexity of DGS at the smoothing parameter. Indeed, all existing algorithms that run in time $2^{n/2+o(n)}$ use exponential space and it is open whether it is possible to sample in subexponential space. If such an algorithm exists, it is not hard to see that all our algorithms will also run in subexponential space on random lattices which would still be a major breakthrough.

Other classes of random lattices In this paper, we studied the Gaussian mass of random *real lattices*. This class enjoys a relatively simple (to use, not to prove!) averaging theorem. Similar results have appeared in [PS26] for random q -ary lattices. A closely related result is available in [KNSW20] which studies matrices where each entry is sampled from an integer Gaussian distribution. Similarly, [CPS⁺20, Appendix A], [LPR13, Section 7] and [SS11, Theorem 2] analyze the case where the entries belong to cyclotomic fields. More recently, some work on random lattices in a fixed genus has led to a new Siegel-like averaging result [vW25] and bounds on the expected value of the Gaussian mass in this setting.

Application to lattice basis reduction algorithms Some lattices used in cryptography are not random in the sense that we use in this paper (e.g. NTRU lattices). However,

our results can still be useful in those settings due to the use of lattice basis reduction algorithms operating on blocks. See for example the analysis of the BKZ algorithm [LN24] which contains a provable bound, and a heuristic bound under the assumption that blocks are “random”.

The results in this paper have been used in [Li25] to show that, under the assumption that local block appearing in the slide reduction algorithms [ALNSD20] are “random”, then there is an exponentially faster provably correct algorithms for δ -SVP for all approximation factors $n^{1/2+\varepsilon} \leq \delta \leq n^{O(1)}$, which is the regime most relevant for cryptography.

Bound on the smoothing parameter for constant ε In this paper, we show (Corollary 3) that for any $\varepsilon > 0$ and almost all lattices $\mathcal{L}$, the smoothing parameter is bounded by

$$\eta_\varepsilon(\mathcal{L}) \leq s_\varepsilon = \left(\frac{\varepsilon + 1 + \sqrt{2\varepsilon + 1}}{\varepsilon^2} \right)^{1/n}. \quad (1)$$

In particular, we obtain that $\eta_{O(1)}(\mathcal{L}) = 1 + o(1)$ as $n \rightarrow \infty$, for most random lattices $\mathcal{L}$.

This should be compared to the existing work. By[‡] a variant[§] of [MR04, Lemma 3.2], it is known that for any lattice $\mathcal{L}$ and any $c > \frac{1}{\sqrt{2\pi}}$,

$$\eta_\varepsilon(\mathcal{L})\lambda_1(\widehat{\mathcal{L}}) \leq c\sqrt{n}$$

where $\varepsilon = (ce^{-\pi c^2}\sqrt{2\pi e})^n(1 + o(1))$. On the other hand, if $\mathcal{L}$ is a random lattice then so is $\widehat{\mathcal{L}}$, so by concentration bounds on λ_1 for random lattices (Theorem 8), it follows that $\lambda_1(\widehat{\mathcal{L}}) \geq \sqrt{\frac{n}{2\pi e}}(1 - o(1))$ and therefore

$$\eta_\varepsilon(\mathcal{L}) \leq c\sqrt{2\pi e}(1 + o(1)).$$

By letting $c \rightarrow \frac{1}{\sqrt{2\pi}}$, one can conclude that $\eta_1(\mathcal{L}) \leq \sqrt{e}$ for most lattices. In other words, it was already implicit in the literature that $\eta_{O(1)}(\mathcal{L}) \leq \sqrt{e}$ and this is the state of the art as far as we know. Therefore our result is a significant improvement compared to the state of the art for random lattices ($\sqrt{e}$ versus $1 + o(1)$). Indeed, even a constant improvement in the smoothing parameter can yield exponential time improvements in the complexity of algorithms. This is for example implicit in [ACKS21] or [ADRS15], and in essence explains why our paper improves upon the state of the art for (approximate) SVP and HSVP on random lattices.

Bound on the smoothing parameter for small ε Looking at (1) when ε becomes sufficiently small, we have $s_\varepsilon \sim 2^{1/n}\varepsilon^{-2/n}$. Combining this with the upper bound (Theorem 8) on $\lambda_1(\mathcal{L})$ we get that for small values of ε ,

$$\eta_\varepsilon(\mathcal{L})\lambda_1(\widehat{\mathcal{L}}) \leq (1 + o(1))\sqrt{\frac{n}{2\pi e}}\varepsilon^{-2/n}. \quad (2)$$

This should be compared with the unconditional result of [ADRS15, Lemma 6.1] that shows that

$$\sqrt{\frac{\log(1/\varepsilon)}{\pi}} < \eta_\varepsilon(\mathcal{L})\lambda_1(\widehat{\mathcal{L}}) < \sqrt{\frac{\beta(\mathcal{L})^2 n}{2\pi e}} \cdot \varepsilon^{-1/n} \cdot (1 + o(1)) \quad (3)$$

where $\beta(\mathcal{L}) \leq 2^{0.401}$ is the generalized kissing number [ACKS21]. It is reasonable to believe that $\beta(\mathcal{L}) \approx 1$ for a random lattice $\mathcal{L}$. If this were the case, then it remains a discrepancy between our bound (2) and the bound (3) of [ADRS15]: $\varepsilon^{-2/n}$ in our case compared to $\varepsilon^{-1/n}$ in theirs. This shows that our upper bound (Corollary 3) is not tight when ε is small. We leave it as future work to improve on this bound.

[‡]The following argument was kindly suggested by a previous reviewer of this article.

[§]Take $c > \frac{1}{\sqrt{2\pi}}$ instead of $c = 1$. This changes C to $C = ce^{-\pi c^2}\sqrt{2\pi e}$.

3 Preliminaries

We denote vectors and matrices in bold case. We denote by $\mathbf{x}^T$ the transpose of the (column) vector $\mathbf{x}$, which is therefore a row vector. For any vector $\mathbf{x} \in \mathbb{R}^n$, we denote by $\|\mathbf{x}\|$ its Euclidean norm. For any $n \geq 1$ and $r > 0$, we denote by B_n the open ball of radius 1 in $\mathbb{R}^n$, and by $B_n(r)$ the open ball of radius r .

For any finite set X , we denote by $\mathcal{U}(X)$ the uniform distribution over X . As usual, if P and Q are two probability distributions over X and Y respectively, we denote by PQ the product distribution over $X \times Y$. For any two distributions P and Q , we denote by $d_{TV}(P, Q)$ the statistical distance (or total variation distance) between P and Q . We say that two distributions P and Q are ε -close if $d_{TV}(P, Q) \leq \varepsilon$.

We denote by ζ the Riemann zeta function, defined for any $s > 1$ by $\zeta(s) = \sum_{k=1}^{\infty} k^{-s}$. Furthermore, it is standard that

$$\zeta(s) - 1 \sim_{s \rightarrow \infty} 2^{-s} \quad (4)$$

3.1 Gamma function

We denote by Γ the gamma function, defined for any $a > 0$ by

$$\Gamma(a) = \int_0^{\infty} t^{a-1} e^{-t} dt.$$

Furthermore, we denote by γ the lower incomplete gamma function, defined for any $a > 0$ and $s \geq 0$ by

$$\gamma(a, x) = \int_0^x t^{a-1} e^{-t} dt.$$

We will make use of the following lower bound for all $a > 0$ and $x \geq 0$:

$$\gamma(a, x) \geq e^{-x} \int_0^x t^{a-1} dt = \frac{x^a e^{-x}}{a}. \quad (5)$$

As well as the following upper bound for all $a > 0$ and $0 \leq x \leq a - 1$:

$$\gamma(a, x) \leq \int_0^x x^{a-1} e^{-x} dt = x^a e^{-x} \quad (6)$$

which holds since $t \mapsto t^{a-1} e^{-t}$ is increasing over $[0, a - 1]$. Finally, we need the following bound on Γ [Rob55]: for $a > 0$,

$$\Gamma(1 + a) \leq \sqrt{2\pi a}^{a+\frac{1}{2}} e^{-a+\frac{1}{12a}}. \quad (7)$$

3.2 Lambert W function

Recall that the Lambert W function is a multivalued function giving the complex solution(s) w to the equation $we^w = z$. In this paper we will only deal with real numbers. It can be shown that for any $x, y \in \mathbb{R}$, the equation

$$ye^y = x$$

can only be solved (for y) if $x \geq -\frac{1}{e}$. For negative numbers $x < 0$, this equation has exactly two solutions $y = W_0(x)$ and $y = W_{-1}(x)$, where W_0 and W_{-1} are the two real branches of the W function. It is known that W_0 is an increasing function while W_{-1} is a decreasing function. Furthermore, for $x \in [-\frac{1}{e}, 0)$, $W_{-1}(x) \leq -1 \leq W_0(x)$. We will also use the fact that the map $x \in [0, 1] \mapsto xe^{1-x}$ is increasing.

3.3 Lattices

A *lattice* $\mathcal{L}$ is a discrete subgroup of $\mathbb{R}^m$. Equivalently it is the set

$$\mathcal{L}(\mathbf{b}_1, \dots, \mathbf{b}_n) = \left\{ \sum_{i=1}^n x_i \mathbf{b}_i : x_i \in \mathbb{Z} \right\}$$

of all integer combinations of n linearly independent vectors $\mathbf{b}_1, \dots, \mathbf{b}_n \in \mathbb{R}^m$. Such $\mathbf{b}_i$'s form a *basis* of $\mathcal{L}$ and are usually collected in matrix form $[\mathbf{b}_1 \ \dots \ \mathbf{b}_n]$. The lattice $\mathcal{L}$ is said to be *full-rank* if $n = m$. We denote by $\lambda_1(\mathcal{L})$ the first minimum of $\mathcal{L}$, defined as the length of a shortest non-zero vector of $\mathcal{L}$. We denote by $\text{vol}(\mathcal{L})$ the volume (or determinant) of $\mathcal{L}$. For a full-rank lattice $\mathcal{L}$, $\text{vol}(\mathcal{L}) = \det(\mathbf{A})$ for any basis $\mathbf{A}$ of $\mathcal{L}$.

For a rank n lattice $\mathcal{L} \subset \mathbb{R}^m$, the *dual lattice*, denoted $\hat{\mathcal{L}}$, is defined as the set of all points in $\text{span}(\mathcal{L})$ that have integer inner products with all lattice points,

$$\hat{\mathcal{L}} = \{\mathbf{w} \in \text{span}(\mathcal{L}) : \forall \mathbf{y} \in \mathcal{L}, \langle \mathbf{w}, \mathbf{y} \rangle \in \mathbb{Z}\}.$$

Similarly, for a lattice basis $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$, we define the dual basis $\mathbf{B}^* = (\mathbf{b}_1^*, \dots, \mathbf{b}_n^*)$ to be the unique set of vectors in $\text{span}(\mathcal{L})$ satisfying $\langle \mathbf{b}_i^*, \mathbf{b}_j \rangle = 1$ if $i = j$, and 0, otherwise. It is easy to show that $\hat{\mathcal{L}}$ is itself a rank n lattice, that $\text{vol}(\hat{\mathcal{L}}) = \frac{1}{\text{vol}(\mathcal{L})}$, and $\mathbf{B}^*$ is a basis of $\hat{\mathcal{L}}$. Given a lattice $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$, we denote $\|\mathbf{B}\|_2 = \max_i \|\mathbf{b}_i\|$. In this paper, all the lattices that we consider will be full rank, *i.e.* $n = m$.

3.4 Lattice problems

We will study the following lattice problems in this paper.

Definition 1. The search problem **SVP** (Shortest Vector Problem) is defined as follows: The input is a basis $\mathbf{B}$ for a lattice $\mathcal{L} \subset \mathbb{R}^n$. The goal is to output a vector $\mathbf{y} \in \mathcal{L}$ with $\|\mathbf{y}\| = \lambda_1(\mathcal{L})$.

Definition 2. For $\gamma = \gamma(n)$, the search problem γ -**SVP** (γ -Approximate Shortest Vector Problem) is defined as follows: The input is a basis $\mathbf{B}$ for a lattice $\mathcal{L} \subset \mathbb{R}^n$. The goal is to output a vector $\mathbf{y} \in \mathcal{L} \setminus \{0\}$ with $\|\mathbf{y}\| \leq \gamma \cdot \lambda_1(\mathcal{L})$.

Definition 3. For $\gamma = \gamma(n)$, the search problem γ -**HSVP** (γ -Hermite Approximate Shortest Vector Problem) is defined as follows: The input is a basis $\mathbf{B}$ for a lattice $\mathcal{L} \subset \mathbb{R}^n$. The goal is to output a vector $\mathbf{y} \in \mathcal{L} \setminus \{0\}$ with $\|\mathbf{y}\| \leq \gamma \cdot \det(\mathcal{L})^{1/n}$.

Definition 4. For $\gamma = \gamma(n)$, the decision problem γ -**GapSVP** (γ -Approximate Gap SVP) is defined as follows: The input is a basis $\mathbf{B}$ for a lattice $\mathcal{L} \subset \mathbb{R}^n$ and a real number $r > 0$. The goal is to accept if $\lambda_1(\mathcal{L}) \leq r$ and reject if $\lambda_1(\mathcal{L}) \geq \gamma r$. Note that this is a promise problem: the program may accept or reject if neither condition holds.

For convenience reasons, when we discuss the running time of the algorithms solving the problems above, we ignore polynomial factors in the bit-length of the individual input basis vectors (*i.e.* we assume the input basis has bit-size polynomial in the ambient dimension n).

3.5 Discrete Gaussian distribution

Let $n \in \mathbb{N}$ and $s > 0$. For any $\mathbf{x} \in \mathbb{R}^n$, we let

$$\rho_s(\mathbf{x}) = e^{-\pi \|\mathbf{x}\|^2 / s^2}.$$

As usual, we extend ρ_s to sets by

$$\rho_s(X) = \sum_{\mathbf{x} \in X} \rho_s(\mathbf{x})$$

for any set X . For any lattice $\mathcal{L} \subset \mathbb{R}^n$, we denote the *discrete Gaussian distribution* over $\mathcal{L}$ by $D_{\mathcal{L},s}(\mathbf{x}) = \frac{\rho_s(\mathbf{x})}{\rho_s(\mathcal{L})}$ for any $\mathbf{x} \in \mathcal{L}$. We denote $D_{\mathcal{L},1}$ by $D_{\mathcal{L}}$ for simplicity. It is well-known by the Poisson summation formula that for any lattice $\mathcal{L}$ and any $s > 0$,

$$\rho_{1/s}(\widehat{\mathcal{L}}) = \text{vol}(\mathcal{L}) s^{-n} \rho_s(\mathcal{L}).$$

See e.g. [Ste17] for a good introduction on this topic. The discrete Gaussian distribution plays an essential role in lattice-based cryptography and an important problem is to be able to sample efficiently from it: this is known as the *discrete Gaussian sampling (DGS)* problem.

Definition 5. For $\delta = \delta(n) \geq 0$, σ a function that maps lattices to non-negative real numbers, and $m = m(n) \in \mathbb{N}$, $\delta\text{-DGS}_{\sigma}^m$ is defined as follows. The input is a basis $\mathbf{B}$ for a lattice $\mathcal{L} \subset \mathbb{R}^n$ and a parameter $s > \sigma(\mathcal{L})$. The goal is to output a sequence of m vectors whose joint distribution is δ -close to m independent samples from $D_{\mathcal{L},s}$.

We omit the parameter δ if $\delta = 0$, and the parameter m if $m = 1$. We stress that δ bounds the statistical distance between the *joint* distribution of the output vectors and m independent samples from $D_{\mathcal{L},s}$.

In general, the smaller s is, the harder it is to construct a sampler for $D_{\mathcal{L},s}$. The notion of smoothing parameter [MR04] captures the idea that sampling for a value of s above this threshold is significantly easier than sampling below because the distribution looks more like a continuous Gaussian. Formally, for any $\varepsilon > 0$, the smoothing parameter of a lattice $\mathcal{L}$ is defined by

$$\eta_{\varepsilon}(\mathcal{L}) = \inf \left\{ s > 0 : \rho_{1/s}(\widehat{\mathcal{L}}) \leq 1 + \varepsilon \right\}.$$

Observe that $\eta_{\varepsilon}(\mathcal{L})$ is a decreasing function of ε .

There are many algorithms to sample above the smoothing parameter [Kle00, GPV08, BLP⁺13], including a time-space trade-off [ACKS21]. Sampling below the smoothing parameter is much more challenging and usually inefficient [ADRS15]. At the extreme, sampling for sufficiently small values of s allows one to solve the Shortest Vector problem (SVP) [ADRS15] which is known to be NP-hard under randomized reduction [Ajt98]. The Monte Carlo Markov Chain based algorithm of [WL19] works for all values of s but the complexity significantly depends on s and the shape of the basis [PS24]. Finally, it should be noted that the problem of estimating the smoothing parameter is provably harder than the bounded distance decoding problem (BDD) and the learning with errors problem (LWE) in certain parameter ranges [CDLP13].

In this paper, we will use the following samplers that works (almost) up to the smoothing parameter and takes time $2^{n/2+o(n)}$ to produce $2^{n/2}$ samples.

Theorem 4 ([ADRS15, Theorem 5.11]). *There is a probabilistic algorithm that, given a lattice $\mathcal{L} \subset \mathbb{R}^n$ and $s \geq \sqrt{2}\eta_{1/2}(\mathcal{L})$ as input, outputs $2^{n/2}$ samples from a distribution $2^{-\Omega(n^2)}$ -close to $D_{\mathcal{L},s}$ in expected time $2^{n/2+o(n)}$.*

4 Random real lattices

Recall that a lattice L is the integer span of a real basis $\mathbf{b}_1, \dots, \mathbf{b}_n$ in $\mathbb{R}^n$. If $\mathbf{B}$ is the matrix whose columns are the $\mathbf{b}_i$, then $L = \mathbf{B}\mathbb{Z}^n$. The classical approach to defining a

probability on the real lattices is the following. First we usually consider lattices modulo scales, so that L and αL are equivalent for any $\alpha \in \mathbb{R}$. Therefore, a lattice is represented by an invertible matrix of determinant 1, that is an element of $\mathrm{SL}_n(\mathbb{R})$. Second, it is clear that many matrices in $\mathrm{SL}_n(\mathbb{R})$ span the same lattice: for instance permuting columns or changing the sign of an even number of columns. In general, matrices $\mathbf{B} \in \mathrm{SL}_n(\mathbb{R})$ and $\mathbf{B}\mathbf{U}$ spans the same lattice for any $\mathbf{U} \in \mathrm{SL}_n(\mathbb{Z})$. The converse is also true and hence we are interested in probability measures on the homogenous space $X_n := \mathrm{SL}_n(\mathbb{R})/\mathrm{SL}_n(\mathbb{Z})$, *i.e.* the set of left cosets of $\mathrm{SL}_n(\mathbb{Z})$ in $\mathrm{SL}_n(\mathbb{R})$.

In this paper, we will identify the set of lattices modulo scaling and the set X_n . This means that we will view an element of X_n either as a lattice or as matrix of determinant of 1, depending on what is more convenient.

4.1 Averaging results of Siegel and Rogers

Let μ be a measure on X_n , Y be a measurable set of lattices in X_n and $\mathbf{B} \in \mathrm{SL}_n(\mathbb{R})$: a “uniform” measure μ should assign the same measure to Y and $\mathbf{B}Y$, *i.e.* it should be (left[¶]) $\mathrm{SL}_n(\mathbb{R})$ –invariant^{||}. Siegel showed [Sie45] that there exists a unique such μ up to a multiplicative factor, and showed that this measure satisfy the following averaging theorem (Theorem 5). We are interested in the unique one which is a probability measure ($\mu(X_n) = 1$) which we denote by μ_n . We also note that the map $X_n \rightarrow X_n, \mathcal{L} \mapsto \hat{\mathcal{L}}$ preserves μ_n so that if $\mathcal{L}$ is distributed according to μ_n then so is its dual $\hat{\mathcal{L}}$.

Theorem 5 (Siegel [Sie45]). *Let $n \geq 1$ and f be a Lebesgue integrable function on $\mathbb{R}^n$ which vanishes everywhere outside a bounded domain, then*

$$\int_{X_n} \sum_{\mathbf{x} \in \mathcal{L} \setminus \{0\}} f(\mathbf{x}) d\mu_n(\mathcal{L}) = \int_{\mathbb{R}^n} f(\mathbf{x}) d\lambda(\mathbf{x}).$$

where λ denotes the usual Lebesgue measure on $\mathbb{R}^n$

This result was later generalized (Theorem 6) by Rogers [Rog55], and the presentation simplified in [MR58] which is probably the most readable reference on the topic. It should be noted that a gap was recently found in the original proof of Rogers but fortunately the result still holds [Kim24].

For any $\ell \leq n$, we say that a matrix $M \in \mathbb{Z}^{n \times \ell}$ is *primitive* if $n - \ell$ columns can be added to it to make up a unimodular matrix. Equivalently, a matrix is primitive if its columns form a primitive set of vectors for $\mathbb{Z}^n$, *i.e.* this set can be extended to form an integer basis of $\mathbb{Z}^n$. Let $\mathrm{PR}_{n,\ell} \subset \mathbb{Z}^{n \times \ell}$ be the set of primitive matrices and $\mathrm{LI}_{n,\ell} \subset \mathbb{Z}^{n \times \ell}$ be the set of matrices whose columns are linearly independent. Macbeath and Rogers’ theorem can be stated as follows. Here, recall again that we identify elements of X_n as both lattices modulo scale, or matrices of determinant 1.

Theorem 6 ([MR58, Theorems 1, 2 and (13)]). *Let $1 \leq \ell \leq n - 1$ and f be Lebesgue integrable on $\mathbb{R}^{n \times \ell}$, then*

$$\begin{aligned} \int_{X_n} \sum_{\mathbf{M} \in \mathrm{LI}_{n,\ell}} f(\mathbf{A}\mathbf{M}) d\mu(\mathbf{A}) &= \zeta(n) \cdots \zeta(n - \ell + 1) \int_{X_n} \sum_{\mathbf{P} \in \mathrm{PR}_{n,\ell}} f(\mathbf{A}\mathbf{P}) d\mu(\mathbf{A}) \\ &= \int_{\mathbb{R}^{n \times \ell}} f(X) d\lambda(X). \end{aligned}$$

[¶]To construct a measure on the homogenous space G/H , it needs to be left-invariant when acting on the left cosets. Equivalently, we could consider the right-invariant measure on the right cosets $G \backslash H$. Then, the lattice spanned by $\mathbf{B}$ would be $\mathbb{Z}^n \mathbf{B}$.

^{||}This notion comes from the theory of Haar measures: the measure is invariant under all group translations $X \in \mathrm{SL}_n(\mathbb{R}) \mapsto \mathbf{B}X$, hence “uniform”.

It is clear that this theorem implies Siegel's theorem when $\ell = 1$ since the set of linearly independent points of $\mathcal{L}$ is exactly $\mathcal{L} \setminus \{0\}$, and it is in fact more general since it works for arbitrary integrable functions. In fact, the result about linearly independent vectors follows easily from the statement about primitive matrices which is the main technical result of [MR58].

4.2 A special case for the variance

Rogers' averaging result is powerful but inconvenient to use for the common case of analyzing the variance of a sum over a lattice. In this paper, we derive a useful corollary for this special case.

Recall that a vector $\mathbf{x} \in \mathbb{Z}^n$ is primitive if and only if $\mathbf{x} \in \text{PR}_{n,1}$. One can check that this is equivalent to saying that $\frac{1}{\alpha}\mathbf{x} \notin \mathbb{Z}^n$ for all integers $\alpha \geq 2$, i.e. $\mathbf{x}$ is not an integer multiple of an integer vector (except by multiplying by ± 1).

Corollary 1. *Let $n \geq 2$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be in L^2 , then*

$$\begin{aligned} \int_{X_n} \sum_{\mathbf{x} \in \mathcal{L}} f(\mathbf{x}) d\mu_n(\mathcal{L}) &= f(0) + \int_{\mathbb{R}^n} f(\mathbf{x}) d\lambda(\mathbf{x}), \\ \int_{X_n} \left(\sum_{\mathbf{x} \in \mathcal{L}} f(\mathbf{x}) \right)^2 d\mu_n(\mathcal{L}) &= \left(\int_{X_n} \sum_{\mathbf{x} \in \mathcal{L}} f(\mathbf{x}) d\mu_n(\mathcal{L}) \right)^2 \\ &\quad + \frac{1}{\zeta(n)} \sum_{\alpha \in \mathbb{N} \setminus \{0\}} \sum_{\beta \in \mathbb{Z} \setminus \{0\}} \int_{\mathbb{R}^n} f(\alpha \mathbf{x}) f(\beta \mathbf{x}) d\lambda(\mathbf{x}). \end{aligned}$$

Proof. The first equality follows directly from Theorem 6. Note that we require Theorem 6 since Theorem 5 was originally only proved for functions with compact support.

$$\int_{X_n} \sum_{\mathbf{x} \in \mathcal{L}} f(\mathbf{x}) d\mu_n(\mathcal{L}) = f(0) + \int_{X_n} \sum_{\mathbf{x} \in \mathcal{L} \setminus \{0\}} f(\mathbf{x}) d\mu_n(\mathcal{L}) = f(0) + \int_{\mathbb{R}^n} f(\mathbf{x}) d\lambda(\mathbf{x}).$$

For the second equality, first observe that

$$\int_{X_n} \left(\sum_{\mathbf{x} \in \mathcal{L}} f(\mathbf{x}) \right)^2 d\mu_n(\mathcal{L}) = \int_{X_n} \sum_{\mathbf{x}, \mathbf{y} \in \mathcal{L}} f(\mathbf{x}) f(\mathbf{y}) d\mu_n(\mathcal{L}).$$

Now let $g : \mathbb{R}^{n \times 2} \rightarrow \mathbb{R}$ be defined by $g(\begin{bmatrix} \mathbf{x} & \mathbf{y} \end{bmatrix}) = f(\mathbf{x})f(\mathbf{y})$ which is clearly Lebesgue integrable. Let $\mathcal{L} \in X_n$ be a lattice and $\mathbf{A}$ be a basis of $\mathcal{L}$. Then

$$\sum_{\mathbf{x}, \mathbf{y} \in \mathcal{L}} f(\mathbf{x}) f(\mathbf{y}) = \sum_{\mathbf{u}, \mathbf{v} \in \mathbb{Z}^n} f(\mathbf{A}\mathbf{u}) f(\mathbf{A}\mathbf{v}) = \sum_{\mathbf{u}, \mathbf{v} \in \mathbb{Z}^n} g(\mathbf{A} \begin{bmatrix} \mathbf{u} & \mathbf{v} \end{bmatrix}).$$

Now there are five cases for $\mathbf{u}$ and $\mathbf{v}$: either $\mathbf{u} = \mathbf{v} = 0$; or $\mathbf{u}$ and $\mathbf{v}$ are linearly independent i.e. $\begin{bmatrix} \mathbf{u} & \mathbf{v} \end{bmatrix} \in \text{LI}_{n,2}$; or $\mathbf{u} = 0$ and $\mathbf{v} \neq 0$; or $\mathbf{u} \neq 0$ and $\mathbf{v} = 0$; or they are linearly dependent and both non zero. The last case is the most interesting: it is not hard to see that if $\mathbf{u}, \mathbf{v} \in \mathbb{Z}^n \setminus \{0\}$ are linearly dependent, then $\mathbf{u} = \alpha \mathbf{p}$ and $\mathbf{v} = \beta \mathbf{p}$ for some unique primitive vector $\mathbf{p} \in \text{PR}_{n,1}$, unique $\alpha \in \mathbb{N} \setminus \{0\}$ and unique $\beta \in \mathbb{Z} \setminus \{0\}$. Since $\mathbf{p}$, α and β are unique, and that conversely the vectors $\alpha \mathbf{p}$ and $\beta \mathbf{p}$ are always linearly dependent and nonzero, there is a bijection between

$$\{(\mathbf{u}, \mathbf{v}) \in (\mathbb{Z}^n \setminus \{0\})^2 : \text{linearly dependent}\}$$

and

$$\{(\alpha \mathbf{p}, \beta \mathbf{p}) : \mathbf{p} \in \text{PR}_{n,1}, \alpha \in \mathbb{N} \setminus \{0\}, \beta \in \mathbb{Z} \setminus \{0\}\}.$$

Therefore,

$$\begin{aligned}
& \sum_{\mathbf{x}, \mathbf{y} \in \mathcal{L}} f(\mathbf{x})f(\mathbf{y}) \\
&= f(0)^2 + \sum_{\mathbf{M} \in \text{LI}_{n,2}} g(\mathbf{AM}) + f(0) \sum_{\mathbf{u} \in \mathbb{Z}^n \setminus \{0\}} f(\mathbf{Au}) + f(0) \sum_{\mathbf{v} \in \mathbb{Z}^n \setminus \{0\}} f(\mathbf{Av}) \\
&\quad + \sum_{\mathbf{p} \in \mathbb{Z}^n \in \text{PR}_{n,1}} \sum_{\alpha \in \mathbb{N} \setminus \{0\}} \sum_{\beta \in \mathbb{Z} \setminus \{0\}} f(\mathbf{A}\alpha\mathbf{p})f(\mathbf{A}\beta\mathbf{p}) \\
&= f(0)^2 + \sum_{\mathbf{M} \in \text{LI}_{n,2}} g(\mathbf{AM}) + 2f(0) \sum_{\mathbf{y} \in \mathcal{L} \setminus \{0\}} f(\mathbf{y}) \\
&\quad + \sum_{\alpha \in \mathbb{N} \setminus \{0\}} \sum_{\beta \in \mathbb{Z} \setminus \{0\}} \sum_{\mathbf{p} \in \mathbb{Z}^n \in \text{PR}_{n,1}} f(\alpha\mathbf{Ap})f(\beta\mathbf{Ap}).
\end{aligned}$$

The integral of the first term is trivial to compute since μ_n is chosen to be a probability measure. We can compute the middle term by Theorem 6:

$$\int_{X_n} f(0) \sum_{\mathbf{y} \in \mathcal{L} \setminus \{0\}} f(\mathbf{y}) d\mu_n(\mathcal{L}) = f(0) \int_{\mathbb{R}^n} f(\mathbf{x}) d\lambda(\mathbf{x}).$$

And the other two by Theorem 6:

$$\begin{aligned}
\int_{X_n} \sum_{\mathbf{M} \in \text{LI}_{n,2}} g(\mathbf{AM}) d\mu_n(\mathbf{A}) &= \int_{\mathbb{R}^{n \times 2}} g(\mathbf{M}) d\lambda(\mathbf{M}) \\
&= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(\mathbf{x})f(\mathbf{y}) d\lambda(\mathbf{x}) d\lambda(\mathbf{y}) \\
&= \left(\int_{\mathbb{R}^n} f(\mathbf{x}) d\lambda(\mathbf{x}) \right)^2
\end{aligned}$$

and

$$\begin{aligned}
\int_{X_n} \sum_{\mathbf{p} \in \mathbb{Z}^n \in \text{PR}_{n,1}} f(\alpha\mathbf{Ap})f(\beta\mathbf{Ap}) d\mu_n(\mathbf{A}) &= \int_{X_n} \sum_{\mathbf{p} \in \text{PR}_{n,1}} f(\alpha\mathbf{Ap})f(\beta\mathbf{Ap}) d\mu_n(\mathbf{A}) \\
&= \frac{1}{\zeta(n)} \int_{\mathbb{R}^n} f(\alpha\mathbf{x})f(\beta\mathbf{x}) d\lambda(\mathbf{x}).
\end{aligned}$$

Therefore,

$$\begin{aligned}
\int_{X_n} \sum_{\mathbf{x}, \mathbf{y} \in \mathcal{L}} f(\mathbf{x})f(\mathbf{y}) d\mu_n(\mathcal{L}) &= f(0)^2 + 2f(0) \int_{\mathbb{R}^n} f(\mathbf{x}) d\lambda(\mathbf{x}) + \left(\int_{\mathbb{R}^n} f(\mathbf{x}) d\lambda(\mathbf{x}) \right)^2 \\
&\quad + \sum_{\alpha \in \mathbb{N} \setminus \{0\}} \sum_{\beta \in \mathbb{Z} \setminus \{0\}} \frac{1}{\zeta(n)} \int_{\mathbb{R}^n} f(\alpha\mathbf{x})f(\beta\mathbf{x}) d\lambda(\mathbf{x})
\end{aligned}$$

which shows the result. $\square$

4.3 Bounds on λ_1

The *Gaussian heuristic* is a well-known approximation used in many works on lattice. This heuristic states that a “random” lattice $\mathcal{L} \subset \mathbb{R}^n$ satisfies that $\lambda_1(\mathcal{L}) \approx \text{vol}(B_n)^{-1/n} \text{vol}(\mathcal{L})^{1/n}$. Note that $\text{vol}(B_n)^{-1/n} \sim \sqrt{\frac{n}{2\pi e}}$ as $n \rightarrow \infty$, so the heuristic

is sometimes stated with $\sqrt{\frac{n}{2\pi e}}$ instead. This heuristic can be made formal for random real lattices, for which the volume is always 1. The most recent result in this vein is the following. There are many ways to prove similar results, see e.g. [Rog56] or the survey [AEN].

Theorem 7 ([LN24, Theorem 6]). *Let $\mathcal{L}$ be a random lattice drawn according to μ_n . Then, with probability at least $1 - o(1)$ as n grows to infinity:*

$$1 - \frac{\log \log n}{n} \leq \frac{\lambda_1(\mathcal{L})}{\text{vol}(B_n)^{-1/n}} \leq 1 + \frac{\log \log n}{n}.$$

In this section, we show that using our averaging result for the special case of the variance, we can recover Theorem 7 but with explicit constants and more flexibility. Although not strictly necessary for this paper, we believe this is an interesting observation which seems to have gone unnoticed in the literature.

First, we start in the classical way by bounding the variance of the number of lattice points in a unit-ball.

Lemma 1. *Let $n \geq 2$ and $r > 0$, then*

$$\begin{aligned} \mathbb{E}_{\mathcal{L} \sim \mu_n} [|\mathcal{L} \cap B_n(r)|] &= 1 + \text{vol}(B_n(r)), \\ \mathbb{V}_{\mathcal{L} \sim \mu_n} [|\mathcal{L} \cap B_n(r)|] &\leq 2A(n) \text{vol}(B_n(r)) \end{aligned}$$

where $A(n) := 1 + \frac{n}{n-1} \frac{\zeta(n-1)-1}{\zeta(n)} = 1 + 2^{1-n}(1 + o(1))$ as $n \rightarrow \infty$.

Proof. Let f_r be the indicator function of the n -dimensional ball $B_n(r)$ of radius r . By Corollary 1, we have that

$$\mu_r := \mathbb{E}_{\mathcal{L} \sim \mu_n} [|\mathcal{L} \cap B_n(r)|] = 1 + \int_{\mathbb{R}^n} f_r(\mathbf{x}) d\lambda(\mathbf{x}) = 1 + \text{vol}(B_n(r)).$$

and

$$\begin{aligned} \sigma_r^2 &:= \mathbb{V}_{\mathcal{L} \sim \mu_n} [|\mathcal{L} \cap B_n(r)|] \\ &= \frac{1}{\zeta(n)} \sum_{\alpha \in \mathbb{N} \setminus \{0\}} \sum_{\beta \in \mathbb{Z} \setminus \{0\}} \int_{\mathbb{R}^n} f_r(\alpha \mathbf{x}) f_r(\beta \mathbf{x}) d\lambda(\mathbf{x}). \end{aligned}$$

But we observe that for any $\alpha \geq 1$ and $\beta \in \mathbb{Z}$, $f_r(\alpha \mathbf{x}) f_r(\beta \mathbf{x}) = f_r(\max(\alpha, |\beta|) \mathbf{x})$. Therefore,

$$\begin{aligned} \sigma_r^2 &= \frac{1}{\zeta(n)} \sum_{\alpha \in \mathbb{N} \setminus \{0\}} \sum_{\beta \in \mathbb{Z} \setminus \{0\}} \int_{\mathbb{R}^n} f_r(\max(\alpha, |\beta|) \mathbf{x}) d\lambda(\mathbf{x}) \\ &= \frac{1}{\zeta(n)} \sum_{\alpha \in \mathbb{N} \setminus \{0\}} \sum_{\beta \in \mathbb{Z} \setminus \{0\}} \max(\alpha, |\beta|)^{-n} \text{vol}(B_n(r)) \\ &= 2 \frac{\text{vol}(B_n(r))}{\zeta(n)} \sum_{\alpha \in \mathbb{N} \setminus \{0\}} \sum_{\beta \in \mathbb{N} \setminus \{0\}} \max(\alpha, \beta)^{-n}. \end{aligned}$$

We now observe that

$$\begin{aligned} &\sum_{\alpha \in \mathbb{N} \setminus \{0\}} \sum_{\beta \in \mathbb{N} \setminus \{0\}} \max(\alpha, \beta)^{-n} \\ &= \sum_{\alpha \in \mathbb{N} \setminus \{0\}} \left(\sum_{\beta=1}^{\alpha} \alpha^{-n} + (\alpha+1)^{-n} + \sum_{\beta=\alpha+2}^{\infty} \beta^{-n} \right) \end{aligned}$$

$$\begin{aligned}
&\leq \sum_{\alpha \in \mathbb{N} \setminus \{0\}} \left(\alpha^{1-n} + (\alpha+1)^{-n} + \int_{\alpha+1}^{\infty} x^{-n} dx \right) \\
&= \zeta(n-1) + \zeta(n) - 1 + \sum_{\alpha \in \mathbb{N} \setminus \{0\}} \frac{(\alpha+1)^{1-n}}{n-1} \\
&= \zeta(n-1) + \zeta(n) - 1 + \frac{\zeta(n-1) - 1}{n-1} \\
&= \zeta(n) + \frac{n}{n-1}(\zeta(n-1) - 1).
\end{aligned}$$

It follows that

$$\sigma_r^2 \leq 2 \operatorname{vol}(B_n(r)) \cdot \left(1 + \frac{n}{n-1} \frac{\zeta(n-1)-1}{\zeta(n)}\right).$$

□

This allows us to bound λ_1 for random lattices. Note that if we take $\alpha = 1 \pm \frac{\log \log n}{n}$ in the following theorem, we recover Theorem 7 with explicit probabilities of $1 - \frac{2 \log(n)}{(\log(n)-2)^2} A(n)$ and $1 - \frac{2 \log(n)^{-1}}{(2 - \log(n)^{-1})^2} A(n)$ respectively for the upper and lower bound, which are indeed $1 - o(1)$, or more precisely $1 - O(\frac{1}{\log(n)})$.

Theorem 8. *Let $n \geq 2$. For any $\alpha > 0$,*

$$\Pr_{\mathcal{L} \sim \mu_n} \left[\lambda_1(\mathcal{L}) \leq \alpha \operatorname{vol}(B_n)^{-1/n} \right] \begin{cases} \leq \frac{2\alpha^n A(n)}{(2-\alpha^n)^2} & \text{if } \alpha < 2^{1/n}, \\ \geq 1 - \frac{2\alpha^n A(n)}{(\alpha^n - 2)^2} & \text{if } \alpha > 2^{1/n}. \end{cases}$$

where $A(n)$ is defined in Lemma 1.

Proof. By Lemma 1, we have that

$$\mu_r := \mathbb{E}_{\mathcal{L} \sim \mu_n} [|\mathcal{L} \cap B_n(r)|] = 1 + \operatorname{vol}(B_n(r))$$

and

$$\sigma_r^2 := \mathbb{V}_{\mathcal{L} \sim \mu_n} [|\mathcal{L} \cap B_n(r)|] \leq 2A(n) \operatorname{vol}(B_n(r)).$$

It follows by Chebyshev's inequality that for any $X > 0$,

$$\Pr_{\mathcal{L}} [||\mathcal{L} \cap B_n(r)| - \mu_r| > X] \leq \frac{\sigma_r^2}{X^2}.$$

We apply the above inequality to study λ_1 . Observe that for any $r > 0$, $\lambda_1(\mathcal{L}) \leq r$ if and only if $|\mathcal{L} \cap B_n(r)| \geq 3$ since as soon as there is a nonzero vector, there are at least three in the ball (a point and its opposite, and the origin). Assume that $\mu_r \leq 3$, i.e. $r^n \operatorname{vol}(B_n) \leq 2$. Then we can let $X = 3 - \mu_r$ and apply the above inequality to get that

$$\begin{aligned}
\Pr_{\mathcal{L}} [\lambda_1(\mathcal{L}) \leq r] &= \Pr_{\mathcal{L}} [|\mathcal{L} \cap B_n(r)| \geq 3] \\
&= \Pr_{\mathcal{L}} [|\mathcal{L} \cap B_n(r)| - \mu_r \geq X] \\
&\leq \Pr_{\mathcal{L}} [||\mathcal{L} \cap B_n(r)| - \mu_r| \geq X] \\
&\leq \frac{\sigma_r^2}{X^2}.
\end{aligned}$$

If we let r_0 be such that $\operatorname{vol}(B_n(r_0)) = 1$ and write $r = \alpha r_0$ then

$$\sigma_r^2 \leq 2 \operatorname{vol}(B_n(\alpha r_0)) A(n) = 2\alpha^n A(n)$$

where $A(n) := 1 + \frac{n}{n-1} \frac{\zeta(n-1)-1}{\zeta(n)}$, and at the same time

$$X = 3 - \mu_r = 2 - \operatorname{vol}(B_n(\alpha r_0)) = 2 - \alpha^n.$$

Finally, we check that the condition $\mu_r \geq 3$ is equivalent to $\alpha^n < 2$.

Conversely, let $Y = \mu_r - 3$ and assume that $Y > 0$, *i.e.* $\alpha^n > 2$. If $||\mathcal{L} \cap B_n(r)| - \mu_r| \leq Y$ then in particular $\mu_r - |\mathcal{L} \cap B_n(r)| \leq Y$ so $|\mathcal{L} \cap B_n(r)| \geq \mu_r - Y = 3$. Hence,

$$\begin{aligned} \Pr_{\mathcal{L}}[\lambda_1(\mathcal{L}) \leq r] &= \Pr_{\mathcal{L}}[|\mathcal{L} \cap B_n(r)| \geq 3] \\ &\geq \Pr_{\mathcal{L}}[||\mathcal{L} \cap B_n(r)| - \mu_r| \leq Y] \\ &= 1 - \Pr_{\mathcal{L}}[||\mathcal{L} \cap B_n(r)| - \mu_r| > Y] \\ &\geq 1 - \frac{\sigma_r^2}{Y^2}. \end{aligned}$$

□

5 On the Gaussian mass of random lattices

In this section, we give a probabilistic estimate of the value of $\rho_s(\mathcal{L})$ when $\mathcal{L}$ is a random real lattice. We derive from this a probabilistic bound on the smoothing parameter of a random lattice. A similar result was shown for “standard” random q -ary lattices (i.i.d. from uniform entries) in [LLBS14, Lemma 3] but only gives the expected value, whereas we also bound the variance. A closely related result is available in [KNSW20] which studies matrices with each entry independently and identically distributed from an integer Gaussian distribution. Similarly, [CPS⁺20, Appendix A], [LPR13, Section 7] and [SS11, Theorem 2] analyzes the Gaussian mass of a random q -ary lattice over cyclotomic fields.

Recall that by random real lattice, we mean $\mathcal{L} \in X_n$ distributed according to μ_n , *i.e.* the Haar measure. See Section 4 for more details. Our first technical result is to obtain the expected value and variance of $\rho_s(\mathcal{L})$ for a random lattice $\mathcal{L}$. We derive from this a probabilistic bound on $\rho_s(\mathcal{L})$ and the smoothing parameter. For reasons that will become clear in the proof of Theorem 9, we prove a more general result on $\rho_s(\mathcal{L} \cap B_n(\ell))$, *i.e.* the Gaussian mass of a lattice restricted to a ball. Note however that if we are only interesting in solving SVP (and not approx-SVP), it would suffice to bound $\rho_s(\mathcal{L})$.

Lemma 2. *For any $n \in \mathbb{N}$, $\ell > 0$ and $s > 0$, let $\mathcal{L} \in X_n$ be distributed according to μ_n . Then*

$$\mathbb{E}_{\mathcal{L}}[\rho_s(\mathcal{L} \cap B_n(\ell))] = 1 + s^n \theta(n, \ell, s), \quad \mathbb{V}_{\mathcal{L}}[\rho_s(\mathcal{L} \cap B_n(\ell))] \leq K(n) s^n \theta(n, \ell \sqrt{2}, s)$$

where $K(n) = \frac{2}{\zeta(n)} \sum_{\alpha, \beta=1}^{\infty} (\alpha^2 + \beta^2)^{-n/2} \leq 2^{1-n/2} (1 + o(1))$ and $\theta(n, \ell, s) = \frac{\gamma(n/2, \pi \ell^2 / s^2)}{\Gamma(n/2)}$ where γ denotes the lower incomplete gamma function. In particular,

$$\mathbb{E}_{\mathcal{L}}[\rho_s(\mathcal{L})] = 1 + s^n, \quad \mathbb{V}_{\mathcal{L}}[\rho_s(\mathcal{L})] \leq K(n) s^n.$$

Proof. Let f_{ℓ} be the indicator function of $B_n(\ell)$. By Corollary 1, we have that

$$\mathbb{E}_{\mathcal{L}}[\rho_s(\mathcal{L} \cap B_n(\ell))] = \rho_s(0) + \int_{\mathbb{R}^n} \rho_s(\mathbf{x}) f_{\ell}(\mathbf{x}) d\lambda(\mathbf{x}).$$

To compute this integral, we perform a polar change of coordinate. Let σ^n denote the spherical measure of the n -sphere S^n . Then

$$\begin{aligned} &\int_{\mathbb{R}^n} \rho_s(\mathbf{x}) f_{\ell}(\mathbf{x}) d\lambda(\mathbf{x}) \\ &= \int_{S^{n-1}} \int_0^{\ell} e^{-\pi \frac{r^2}{s^2}} r^{n-1} dr d\sigma^{n-1}(\omega) \\ &= \sigma^{n-1}(S^{n-1}) \int_0^{\ell} e^{-\pi \frac{r^2}{s^2}} r^{n-1} dr \end{aligned}$$

$$\begin{aligned}
&= \frac{2\pi^{n/2}}{\Gamma(\frac{n}{2})} \int_0^{\pi \frac{\ell^2}{s^2}} e^{-t} \left(s \sqrt{\frac{t}{\pi}} \right)^{n-2} \frac{s^2}{2\pi} dt \quad \text{by the change } t = \pi \frac{r^2}{s^2} \\
&= \frac{s^n}{\Gamma(\frac{n}{2})} \int_0^{\pi \frac{\ell^2}{s^2}} e^{-t} t^{\frac{n}{2}-1} dt \\
&= s^n \frac{\gamma(\frac{n}{2}, \pi \frac{\ell^2}{s^2})}{\Gamma(\frac{n}{2})}.
\end{aligned}$$

By Corollary 1, and a similar integral calculation, we also have that

$$\begin{aligned}
&\mathbb{V}_{\mathcal{L} \sim \mu_n}[\rho_s(\mathcal{L} \cap B_n(\ell))] \\
&= \frac{1}{\zeta(n)} \sum_{\alpha \in \mathbb{N} \setminus \{0\}} \sum_{\beta \in \mathbb{Z} \setminus \{0\}} \int_{\mathbb{R}^n} \rho_s(\alpha \mathbf{x}) \rho_s(\beta \mathbf{x}) f_\ell(\alpha \mathbf{x}) f_r(\beta \mathbf{x}) d\lambda(\mathbf{x}) \\
&= \frac{1}{\zeta(n)} \sum_{\alpha \in \mathbb{N} \setminus \{0\}} \sum_{\beta \in \mathbb{Z} \setminus \{0\}} \int_{\mathbb{R}^n} \rho_{s/\sqrt{\alpha^2 + \beta^2}}(\mathbf{x}) f_\ell(\max(\alpha, |\beta|)\mathbf{x}) d\lambda(\mathbf{x}) \\
&= 2 \frac{s^n}{\zeta(n)} \sum_{\alpha, \beta=1}^{\infty} (\alpha^2 + \beta^2)^{-n/2} \frac{\gamma\left(\frac{n}{2}, \pi \frac{(\alpha^2 + \beta^2)\ell^2}{s^2 \max(\alpha^2, \beta^2)}\right)}{\Gamma(\frac{n}{2})} \\
&\leq 2 \frac{s^n}{\zeta(n)} \sum_{\alpha, \beta=1}^{\infty} (\alpha^2 + \beta^2)^{-n/2} \frac{\gamma\left(\frac{n}{2}, 2\pi \frac{\ell^2}{s^2}\right)}{\Gamma(\frac{n}{2})} \quad \text{since } \gamma(\frac{n}{2}, \cdot) \text{ is increasing} \\
&= 2 \frac{s^n}{\zeta(n)} \frac{\gamma(\frac{n}{2}, 2\pi \frac{\ell^2}{s^2})}{\Gamma(\frac{n}{2})} \sum_{\alpha, \beta=1}^{\infty} (\alpha^2 + \beta^2)^{-n/2}.
\end{aligned}$$

Furthermore,

$$\begin{aligned}
\sum_{\alpha, \beta=1}^{\infty} (\alpha^2 + \beta^2)^{-n/2} &= \sum_{\alpha=1}^{\infty} \left((\alpha^2 + 1)^{-n/2} + \sum_{\beta=2}^{\infty} (\alpha^2 + \beta^2)^{-n/2} \right) \\
&\leq \sum_{\alpha=1}^{\infty} \left((\alpha + 1)^{-n/2} + \sum_{\beta=2}^{\infty} (\alpha^2 + \beta^2)^{-n/2} \right) \\
&\leq \zeta(\frac{n}{2}) - 1 + \sum_{\alpha=1}^{\infty} \int_1^{\infty} (\alpha + \beta)^{-n/2} d\beta \\
&= \zeta(\frac{n}{2}) - 1 + \sum_{\alpha=1}^{\infty} 2 \frac{(\alpha+1)^{1-\frac{n}{2}}}{n-2} \\
&= \zeta(\frac{n}{2}) - 1 + \frac{2}{n-2} (\zeta(\frac{n}{2}) - 1) \\
&= 2^{-n/2} (1 + o(1))
\end{aligned}$$

as $n \rightarrow \infty$. The final part of the result follows by letting $\ell \rightarrow \infty$ and noting that $\theta(n, \ell, s) \rightarrow 1$ since $\Gamma(\frac{n}{2}) = \lim_{x \rightarrow \infty} \gamma(\frac{n}{2}, x)$ by definition. $\square$

Corollary 2. For any $n \in \mathbb{N}$, $s > 0$, $\ell > 0$ and $\alpha > 0$,

$$\Pr_{\mathcal{L} \sim \mu_n}[|\rho_s(\mathcal{L} \cap B_n(\ell)) - 1 - s^n \theta(n, \ell, s)| > \alpha] \leq \frac{K(n) s^n \theta(n, \ell \sqrt{2}, s)}{\alpha^2}$$

where θ and K are defined in Lemma 2. In particular,

$$\Pr_{\mathcal{L} \sim \mu_n}[|\rho_s(\mathcal{L}) - 1 - s^n| > \alpha] \leq \frac{K(n) s^n}{\alpha^2}.$$

Symbol	Definition	Defined in	Upper bound
$A(n)$	$1 + \frac{n}{n-1} \frac{\zeta(n-1) - 1}{\zeta(n)}$	Lemma 1	$1 + 2^{1-n}(1 + o(1))$
$K(n)$	$\frac{2}{\zeta(n)} \sum_{\alpha, \beta=1}^{\infty} (\alpha^2 + \beta^2)^{-n/2}$	Lemma 2	$2^{1-n/2}(1 + o(1))$
$\theta(n, \ell, s)$	$\frac{\gamma(n/2, \pi \ell^2 / s^2)}{\Gamma(n/2)}$	Lemma 2	

Figure 2: Summary of the important quantities appearing in Section 5.

Proof. This is a direct application of Lemma 2 and Chebyshev's inequality. $\square$

The previous lemma allows us to derive a probabilistic bound on the smoothing parameter $\eta_\varepsilon(\mathcal{L})$ of a random lattice $\mathcal{L}$.

Corollary 3. *For any $n \in \mathbb{N}$ and $\varepsilon > 0$, let $s_\varepsilon = \left(\frac{\varepsilon+1+\sqrt{2\varepsilon+1}}{\varepsilon^2} \right)^{1/n}$. Then*

$$\Pr_{\mathcal{L} \sim \mu_n} [\eta_\varepsilon(\mathcal{L}) > s_\varepsilon] \leq \frac{K(n)}{2}$$

where K is defined in Lemma 2.

Proof. Recall that if $\mathcal{L} \in X_n$ is distributed according to μ_n then its dual $\widehat{\mathcal{L}}$ is also distributed according to μ_n . Let $\alpha = \varepsilon - s_\varepsilon^{-n}$ and check that $\alpha > 0$. We can therefore apply Corollary 2 to get that

$$\begin{aligned} \Pr_{\mathcal{L}} [\rho_{1/s_\varepsilon}(\widehat{\mathcal{L}}) > 1 + \varepsilon] &= \Pr_{\mathcal{L}} [\rho_{1/s_\varepsilon}(\widehat{\mathcal{L}}) - 1 - s_\varepsilon^{-n} > \alpha] \\ &\leq \Pr_{\mathcal{L}} [|\rho_{1/s_\varepsilon}(\widehat{\mathcal{L}}) - 1 - s_\varepsilon^{-n}| > \alpha] \\ &\leq \frac{K(n)}{s_\varepsilon^n \alpha^2}. \end{aligned}$$

A routine calculation shows that** $s_\varepsilon^n \alpha^2 = s_\varepsilon^n (\varepsilon - s_\varepsilon^{-n})^2 = 2$. Furthermore, for any lattice $\mathcal{L}$, if $\rho_{1/s_\varepsilon}(\widehat{\mathcal{L}}) \leq 1 + \varepsilon$ then $\eta_\varepsilon(\mathcal{L}) \leq s_\varepsilon$. Therefore,

$$\Pr_{\mathcal{L}} [\eta_\varepsilon(\mathcal{L}) > s_\varepsilon] \leq \Pr_{\mathcal{L}} [\rho_{1/s_\varepsilon}(\widehat{\mathcal{L}}) > 1 + \varepsilon] \leq \frac{K(n)}{2}. \quad (8)$$

$\square$

The quantities $A(n)$, $K(n)$ and $\theta(n, \ell, s)$ will play an important role in the next section. Figure 2 summarizes the definitions and bounds of these quantities.

6 Application to the Hermite and approximate SVP

6.1 Hermite SVP

In this section, we use our probabilistic bound on the smoothing parameter $\eta_\varepsilon(\mathcal{L})$ to solve the approximate (Hermite) SVP for random lattices. Our algorithm is conceptually simple:

**Technically, there is a second choice $\tilde{s}_\varepsilon = \left(\frac{\varepsilon+1-\sqrt{2\varepsilon+1}}{\varepsilon^2} \right)^{1/n}$ which satisfies that $\tilde{s}_\varepsilon^n (\varepsilon - \tilde{s}_\varepsilon^{-n})^2 = 2$.

However note that $\tilde{s}_\varepsilon \leq \varepsilon^{-1/n}$ so if we applied Corollary 2, we would get the probability that $\rho_{1/\tilde{s}_\varepsilon}(\widehat{\mathcal{L}}) > 1 + \tilde{s}_\varepsilon^{-n} + \alpha$ for some positive $\alpha > 0$, but since $1 + \tilde{s}_\varepsilon^{-n} > 1 + \varepsilon$, this would not help us bound the smoothing parameter η_ε .

we sample a large number of vectors from $D_{\mathcal{L},s}$ for $s = \sqrt{2}\eta_{1/2}(\mathcal{L})$ and we return the shortest nonzero vector among them. The trade-off lies in the number of samples: the more we have, the more likely we are to find a short vector but the more expensive the algorithm becomes. The sampler that we used (Theorem 4) runs in time $2^{n/2+o(n)}$ but gives us $2^{n/2+o(n)}$ vectors at a time. Surprisingly, this already allows us to solve 1.123-SVP, but by calling the sampler an exponential number of times, we can decrease the approximation factor to be arbitrarily close to 1.

Theorem 9. *There is a randomized algorithm that for every $n \geq 2$ and $\beta \in (0, \frac{n-2}{2n})$ and on a fraction at least $1 - 2^{1-n/2}(1+o(1)) - (2e\beta)^{-n/2}O(n^{3/2})$ of random lattices $\mathcal{L}$ according to μ_n , outputs in time $e^{o(n)}(\beta e^{1-\beta})^{-n/2} + 2^{n/2+o(n)}$ and space $2^{n/2+o(n)}$ a nonzero vector of $\mathcal{L}$ of length at most $s_{1/2}\sqrt{\frac{n\beta}{\pi}}$ with probability at least $1/2$, where $s_{1/2} = (6+4\sqrt{2})^{1/n} = 1+o(1)$ is defined in Corollary 3.*

Remark 1. Although this theorem can be applied up to $\beta \approx \frac{1}{2}$, values above $\beta_{\max} = -W_0(-\frac{1}{2e}) \approx 0.2319609530$ are not interesting. Indeed, for $\beta = \beta_{\max}$, the time complexity of the algorithm is exactly $2^{n/2+o(n)}$ and it cannot become any smaller by increasing β . By taking $\beta \leq \beta_{\max}$, we also guarantee that $\beta < \frac{1}{e}$ and therefore $2^{1-n/2+o(n)} = O((2e\beta)^{-n/2})$, so the fraction of lattices on which the algorithm works is at least $1 - (2e\beta)^{-n/2}O(n^{-3/2})$.

Similarly, this theorem is not useful for $\beta < \frac{1}{2e}$ since for such values the theorem cannot guarantee that the algorithm works on a positive fraction of lattices.

Proof. See Figure 2 for a summary of the quantities $A(n)$, $K(n)$ and $\theta(n, \ell, s)$ appearing in the proof. Let $s_{1/2}$ be given by Corollary 3, $\sigma = \sqrt{2}s_{1/2}$, $\ell = s_{1/2}\sqrt{\frac{n\beta}{\pi}}$ and N to be fixed later. Note that $\sigma^n \geq \sqrt{2}^n \geq 2$ since $n \geq 2$ and $s_{1/2} \geq 1$. Consider the following algorithm:

- Sample N vectors independently according to $D_{\mathcal{L},\sigma}$ by Theorem 4.
- Return the shortest nonzero vector.

We will analyze this algorithm. First the running time is clear by Theorem 4: the algorithm takes time $(N + 2^{n/2}) \cdot 2^{o(n)}$. Next we observe that by Corollary 3 for $\varepsilon = 1/2$, and with probability at least $1 - \frac{K(n)}{2} = 1 - 2^{-n/2}(1+o(n))$ over the choice of $\mathcal{L}$ we have

$$\eta_{1/2}(\mathcal{L}) \leq s_{1/2}. \quad (9)$$

Furthermore, by Corollary 2 for $\alpha = \frac{1}{2}\sigma^n\theta(n, \ell, \sigma)$, and with probability at least $1 - \frac{1}{X}$ where $X = 2^{n/2-3}\sigma^n \frac{\theta(n, \ell, \sigma)^2}{\theta(n, \ell\sqrt{2}, \sigma)}(1+o(n))$ over the choice of $\mathcal{L}$ we have

$$|\rho_\sigma(\mathcal{L} \cap B_n(\ell)) - 1 - \sigma^n\theta(n, \ell, \sigma)| < \frac{1}{2}\sigma^n\theta(n, \ell, \sigma). \quad (10)$$

Finally, by the same argument, and with probability at least $1 - 2^{-n/2}(1+o(n))$ over the choice of $\mathcal{L}$ we have

$$|\rho_\sigma(\mathcal{L}) - 1 - \sigma^n| < \sqrt{2\sigma^n}. \quad (11)$$

By a union bound, we have that (9), (10) and (11) hold at the same time for a fraction at least $1 - 2^{1-n/2}(1+o(1)) - \frac{1}{X}$ of random lattices. Assume that we are in this case. It will be useful to note that $s_{1/2} = (6+4\sqrt{2})^{1/n}$. Therefore

$$11^{1/n} \leq s_{1/2} \leq 12^{1/n}.$$

Since $\sqrt{2}\eta_{1/2}(\mathcal{L}) \leq \sqrt{2}s_{1/2} = \sigma$ we can indeed apply Theorem 4 to sample from $D_{\mathcal{L},\sigma}$. Therefore, the probability that each sample is nonzero and of length at most ℓ is

$$p := \frac{\rho_\sigma(\mathcal{L} \setminus \{0\} \cap B_n(\ell))}{\rho_\sigma(\mathcal{L})} \geq \frac{\frac{1}{2}\sigma^n\theta(n, \ell, \sigma)}{1 + \sigma^n + \sqrt{2\sigma^n}} \quad \text{by (10) and (11)}$$

$$\geq \frac{\frac{1}{2}\sigma^n\theta(n, \ell, \sigma)}{2\sigma^n} = \frac{\theta(n, \ell, \sigma)}{4} \quad \text{since } \sigma^n \geq 2.$$

Therefore if $N = \Omega(1/p)$, then with constant probability we will get a vector in $(\mathcal{L} \setminus \{0\}) \cap B_n(\ell)$.

We now need to obtain a lower bound on $\theta(n, \ell, \sigma)$. Check that $\pi\ell^2/\sigma^2 = \frac{n}{2}\beta$. Let $a = \frac{n}{2}$ and then

$$\begin{aligned} \theta(n, \ell, \sigma) &= \frac{\gamma(n/2, \pi\ell^2/\sigma^2)}{\Gamma(n/2)} \\ &= \frac{\gamma(a, a\beta)}{\Gamma(a)} \geq \frac{\gamma(a, a\beta)}{\Gamma(a+1)} \\ &\geq \frac{(a\beta)^a e^{-a\beta}}{a \cdot \sqrt{2\pi} a^{a+\frac{1}{2}} e^{-a+\frac{1}{12a}}} && \text{by (5) and (7)} \\ &= \frac{1}{a^{3/2}\sqrt{2\pi}} \beta^a e^{a-a\beta-\frac{1}{12a}} \\ &= (\beta e^{1-\beta})^{n/2} \Omega(n^{-3/2}). \end{aligned}$$

It follows that

$$1/p = (\beta e^{1-\beta})^{-n/2} \frac{1}{\Omega(n^{-3/2})} = e^{o(n)} (\beta e^{1-\beta})^{-n/2}$$

which shows the result for the time complexity. Recall that the result holds for a fraction at least $1 - 2^{1-n/2}(1 + o(1)) - \frac{1}{X}$ of lattices, where

$$\begin{aligned} X &= 2^{n/2-3}\sigma^n \frac{\theta(n, \ell, \sigma)^2}{\theta(n, \ell\sqrt{2}, \sigma)} (1 + o(n)) \\ &= 2^{n/2-3}\sigma^n \frac{\gamma(a, a\beta)^2}{\Gamma(a)\gamma(a, 2a\beta)} (1 + o(n)) \quad \text{with the same notation as above} \\ &\geq 2^{n/2-3} \cdot 11\sqrt{2}^n \cdot \frac{\frac{(a\beta)^{2a} e^{-2a\beta}}{a^2}}{\sqrt{2\pi} a^{a+\frac{1}{2}} e^{-a+\frac{1}{12a}} (2a\beta)^a e^{-2a\beta}} && \begin{array}{l} \text{by (5), (6) and (7)} \\ \text{since } 2a\beta \leq a-1 \end{array} \\ &= \frac{11}{8} 2^n \frac{(a\beta)^a}{a^{a+\frac{3}{2}} e^{-a+\frac{1}{12a}} 2^a} \\ &= (2e\beta)^{n/2} \Omega(n^{-3/2}) && \text{since } a = \frac{n}{2}. \end{aligned}$$

□

The previous result allows us to show that we have an algorithm that returns relatively short vectors on average but note that the bound does not depend on the lattice (more precisely, it is related to the volume of the lattice because our random lattices are scaled to have volume 1). This is known as the α -Hermite SVP (HSVP). The following corollary is a simplified version of Theorem 9.

Corollary 4 (Hermite SVP). *There is a randomized algorithm that for every $n \geq 2$ and $\beta \in [\frac{1}{2e}, \beta_{\max}]$, where $\beta_{\max}$ is defined in Remark 1, solves $(1 + o(1))\sqrt{\frac{n\beta}{\pi}}$ -HSVP in time $e^{o(n)}(\beta e^{1-\beta})^{-n/2}$ and space $2^{n/2+o(n)}$ with probability at least $1/2$ on a fraction at least $1 - e^{o(n)}(2e\beta)^{-n/2}$ of random lattices according to μ_n .*

Proof. This is a direct consequence of Theorem 9 since $s_{1/2} \leq 12^{1/n} = 1 + o(1)$. The only thing to note is that the time complexity is

$$e^{o(n)}(\beta e^{1-\beta})^{-n/2} + 2^{n/2+o(n)}$$

but by Remark 1, the first term is always larger than the second when $\beta \leq \beta_{\max}$. Similarly we use Remark 1 to simplify the fraction of lattices. □

6.2 Approximate SVP

The more common γ -SVP problem asks to relate the length of the vectors to the first minimum $\lambda_1(\mathcal{L})$. To do so, we rely on a probabilistic lower bound on λ_1 for random lattices and use the results of the previous section on the Hermite SVP.

Theorem 10 (γ -SVP). *There is a randomized algorithm that for every $n \geq 1$, $\gamma \in (1, \sqrt{2e\beta_{\max}}]$ and $\beta \in [\frac{1}{2e}, \frac{\gamma^2}{2e})$, where $\beta_{\max}$ is defined in Remark 1, solves γ -SVP in time $e^{o(n)}(\beta e^{1-\beta})^{-n/2}$ and space $2^{n/2+o(n)}$ with probability at least $1/2$ on a fraction at least $1 - O(n^{3/2})(2e\beta)^{-n/2} - O(n^{-1/2}) \left(\frac{2e\beta}{\gamma^2}\right)^{n/2}$ of random lattices $\mathcal{L}$ according to μ_n .*

Remark 2. If we let $\beta \rightarrow \frac{\gamma^2}{2e}$ then the complexity of the algorithm will tend to $e^{o(n)}(\frac{2}{\gamma^2}e^{\gamma^2/2e})^{n/2}$, but the fraction of random lattices on which the algorithm works is unclear. This limit behaviour is analyzed in Corollary 5 and Corollary 6. A similar problem occurs if we let $\gamma \rightarrow 1$, which is analyzed specifically in Section 6.3.

Proof. Let $\beta \in (\frac{1}{2e}, \beta_{\max})$ to be fixed later where $\beta_{\max}$ is defined in Remark 1. We apply Theorem 9 to get an algorithm that returns a nonzero vector on a lattice $\mathcal{L}$ with probability at least $1/2$ and of length at most $\ell = s_{1/2} \sqrt{\frac{n\beta}{\pi}}$, where $s_{1/2} = (6 + 4\sqrt{2})^{1/n} \leq 12^{1/n}$. This algorithm works on a fraction at least $1 - 2^{1-n/2}(1 + o(n)) - (2e\beta)^{-n/2}O(n^{3/2})$ of random lattices.

Let $\alpha = \frac{\ell}{\gamma} \text{vol}(B_n)^{1/n}$. Assuming for now that $\alpha < 2^{1/n}$, we apply Theorem 8 to get that for a fraction at least $1 - \varepsilon$ of lattices, where $\varepsilon = \frac{2\alpha^n(1+o(1))}{(2-\alpha^n)^2}$, we have $\lambda_1(\mathcal{L}) \geq \alpha \text{vol}(B_n)^{-1/n}$. Therefore, for a fraction at least $1 - \varepsilon - (2e\beta)^{-n/2}O(n^{3/2}) - 2^{-n/2}(1 + o(1))$ of lattices, the nonzero vector returned by the algorithm is of length at most

$$\ell = \gamma \alpha \text{vol}(B_n)^{-1/n} \leq \gamma \lambda_1(\mathcal{L}).$$

Recall that we need to satisfy the constraint $\alpha < 2^{1/n}$. Check that

$$\begin{aligned} \alpha^n &\leq 12 \left(\frac{n\beta}{\pi\gamma^2} \right)^{n/2} \text{vol}(B_n) \\ &= 12 \left(\frac{n\beta}{\pi\gamma^2} \right)^{n/2} O(n^{-1/2}) \left(\frac{2\pi e}{n} \right)^{n/2} \\ &= O(n^{-1/2}) \left(\frac{2e\beta}{\gamma^2} \right)^{n/2}. \end{aligned}$$

In particular, if $2e\beta < \gamma^2$ then $\alpha \rightarrow 0$ as $n \rightarrow \infty$ so the condition $\alpha < 2^{1/n}$ is clearly satisfied for large enough n . If we further restrict $\gamma \leq \gamma_{\max}$ so that $\gamma_{\max}^2/2e = \beta_{\max}$, then by Remark 1 we can simplify the time complexity as in the proof of Corollary 4. $\square$

The previous theorem is difficult to use because of the parameter β which balances the complexity and the fraction of lattices on which the algorithm succeeds. The following corollaries are simplified versions for hardcoded fraction of lattices.

Corollary 5 (γ -SVP for almost all lattices). *There is a randomized algorithm that for every $\gamma \in (\frac{1}{0.99}, \sqrt{2e\beta_{\max}}]$ and $n \geq 1$ solves γ -SVP in time $(0.1821\gamma^2 e^{1-0.1821\gamma^2})^{-n/2} e^{o(n)}$ and space $2^{n/2+o(n)}$ with probability at least $1/2$ on a fraction at least $1 - 0.99^{n/2+o(n)}$ of random lattices $\mathcal{L}$ according to μ_n .*

Proof. Apply Theorem 10 with $\beta = 0.99\frac{\gamma^2}{2e}$. Then the fraction of lattices on which the algorithm works is $1 - e^{o(n)}(2e\beta)^{-n/2} - e^{o(n)}\left(\frac{2e\beta}{\gamma^2}\right)^{n/2}$. The last term is clearly equal to $e^{o(n)}0.99^{n/2}$. The first term is bounded by $e^{o(n)}0.99^{n/2}$ when $\gamma \geq \frac{1}{0.99}$. Finally, substituting into the runtime complexity gives the result. $\square$

Corollary 6 (γ -SVP for most lattices). *For every $k > 0$, there is a randomized algorithm that for every $\gamma \in (1, \sqrt{2e\beta_{\max}}]$ and $n \geq -\frac{k}{\ln \gamma}W_{-1}(-\frac{\ln \gamma}{k})$ solves γ -SVP in space $2^{n/2+o(n)}$ and time $e^{o(n)}(\frac{2}{\gamma^2}e^{\gamma^2/2e})^{n/2}$ with probability at least $1/2$ on a fraction at least $1 - O(n^{-k})$ of random lattices $\mathcal{L}$ according to μ_n .*

Proof. Fix $k > 0$ and let $n \geq 1$ and γ as specified. Let $\beta = \frac{\gamma^2}{2e}n^{-2k/n}$ and note that $\beta \leq \frac{\gamma^2}{2e}$. Furthermore, a rather tedious computation shows that

$$\beta \geq \frac{1}{2e} \quad \text{whenever} \quad n \geq -\frac{k}{\ln \gamma}W_{-1}(-\frac{\ln \gamma}{k}).$$

We can therefore apply Theorem 10 to get an algorithm which solves γ -SVP. The fraction of lattices on which the algorithm works is

$$\begin{aligned} & 1 - O(n^{3/2})(2e\beta)^{-n/2} - O(n^{-1/2})\left(\frac{2e\beta}{\gamma^2}\right)^{n/2} \\ &= 1 - O(n^{3/2})\left(\gamma^2 n^{-2k/n}\right)^{-n/2} - O(n^{-1/2})\left(n^{-2k/n}\right)^{n/2} \\ &= 1 - O(n^{k+3/2})\gamma^{-n} - O(n^{-k-1/2}) \\ &= 1 - O(n^{-k-1/2}). \end{aligned}$$

The time complexity of the algorithm is

$$\begin{aligned} e^{o(n)}(\beta e^{1-\beta})^{-n/2} &= e^{o(n)}\left(\frac{\gamma^2}{2e}n^{-2k/n}e^{1-\frac{\gamma^2}{2e}n^{-2k/n}}\right)^{-n/2} \\ &= e^{o(n)}n^k\left(\frac{\gamma^2}{2}e^{-\frac{\gamma^2}{2e}n^{-2k/n}}\right)^{-n/2} \\ &= e^{o(n)}\left(\frac{2}{\gamma^2}e^{\frac{\gamma^2}{2e}\left(1+O\left(\frac{\log n}{n}\right)\right)}\right)^{n/2} \\ &= e^{o(n)}e^{\frac{n}{2}O\left(\frac{\log n}{n}\right)}\left(\frac{2}{\gamma^2}e^{\frac{\gamma^2}{2e}}\right)^{n/2} \\ &= e^{o(n)}\left(\frac{2}{\gamma^2}e^{\frac{\gamma^2}{2e}}\right)^{n/2} \end{aligned}$$

which shows the result. $\square$

6.3 Exact SVP

The results of the previous section allow us to solve γ -SVP for γ arbitrarily close to 1. Unfortunately, the (lower bound on the) fraction of lattices on which they apply tends to 0 as $\gamma \rightarrow 1$. This leaves open whether the algorithm can solve SVP. In the next theorem, we show that this is the case. Even though the algorithm is the same, the analysis is quite different but also simpler because it only relies on the probabilistic bounds on $\rho_s(\mathcal{L})$.

Theorem 11 (SVP). *There is a randomized algorithm that for every $n \geq 1$ and $\alpha > 2^{1/n}$, and on a fraction at least $1 - 2^{1-n/2}(1 + o(1)) - \frac{2\alpha^n A(n)}{(\alpha^n - 2)^2}$ of random lattices $\mathcal{L}$ according to μ_n , outputs in time $2^{\tau n + o(n)}$ and space $2^{n/2 + o(n)}$ a shortest nonzero vector of $\mathcal{L}$ with probability at least $1/2$, where $\tau = \frac{1}{2} + \frac{\alpha^2}{4e \ln 2}$ and $A(n)$ is defined in Theorem 8.*

Remark 3. If we let $n \rightarrow \infty$ and $\alpha \rightarrow 1$ then we can solve SVP in time $2^{\left(\frac{1}{2} + \frac{\alpha^2}{4e \ln 2}\right)n + o(n)}$, however the fraction of lattices on which the algorithm works is unclear. This is formalised in Corollary 8 which solves SVP on “most” $(1 - 1/\text{poly}(n))$ lattices. If we want to solve SVP for “almost all” $(1 - e^{-\Omega(n)})$ lattices, we need to make the complexity slightly worse to obtain a better fraction, as in Corollary 7.

Proof. Let $s_{1/2}$ be given by Corollary 3, $\sigma = \sqrt{2}s_{1/2}$, and N to be fixed later. Consider the following algorithm:

- Sample N vectors independently according to $D_{\mathcal{L}, \sigma}$ by Theorem 4.
- Return the shortest nonzero vector.

We will analyze this algorithm. First the running time is clear by Theorem 4: the algorithm takes time $N + 2^{n/2 + o(n)}$. Next we observe that by Corollary 2 for $\alpha = \sqrt{2}\sigma^n$ and Corollary 3 for $\varepsilon = 1/2$, and with probability at least $1 - 2^{1-n/2}(1 + o(n))$ we have both

$$\rho_\sigma(\mathcal{L}) \leq 1 + \sigma^n + \sqrt{2}\sigma^n \quad \text{and} \quad \eta_{1/2}(\mathcal{L}) \leq s_{1/2}.$$

Assume that we are in this case. It will be useful to note that $s_{1/2} = (6 + 4\sqrt{2})^{1/n}$. Therefore $s_{1/2} \leq 12^{1/n}$ and $\rho_\sigma(\mathcal{L}) \leq 2^{n/2 + o(n)}$. Since $\sqrt{2}\eta_{1/2}(\mathcal{L}) \leq \sqrt{2}s_{1/2} = \sigma$ we can indeed apply Theorem 4 to sample from $D_{\mathcal{L}, \sigma}$. Check that

$$p := \Pr_{\mathbf{x} \sim D_{\mathcal{L}, \sigma}}[\|\mathbf{x}\| = \lambda_1(\mathcal{L})] \geq \frac{e^{-\pi \frac{\lambda_1(\mathcal{L})^2}{\sigma^2}}}{\rho_\sigma(\mathcal{L})}.$$

Let $\alpha > 2^{1/n}$. Apply Theorem 8 to get that for a fraction at least $1 - \delta$ of lattices $\mathcal{L}$, where $\delta = \frac{2\alpha^n A(n)}{(\alpha^n - 2)^2}$, we have $\lambda_1(\mathcal{L}) \leq \alpha \text{vol}(B_n)^{-1/n}$. By Stirling’s formula, $\text{vol}(B_n)^{-1/n} = \frac{1}{\sqrt{\pi}} \Gamma(\frac{n}{2} + 1)^{1/n} = \sqrt{\frac{n}{2\pi e}}(1 + o(1))$. Therefore, by putting everything together,

$$p \geq \frac{\exp\left(-\pi \alpha^2 \frac{n}{2e} (1 + o(1))^2 \frac{1}{2s_{1/2}^2}\right)}{2^{n/2 + o(n)}} \geq 2^{n/2} e^{-\alpha^2 \frac{n}{4e} + o(n)} = 2^{\tau n + o(n)}$$

where $\tau = \frac{1}{2} + \frac{\alpha^2}{4e \ln 2}$. Note that this probability only depends on the particular choice of $\mathcal{L}$ and holds for a fraction at least $1 - \delta - 2^{1-n/2}(1 + o(n))$ of random lattices. As a result,

$$\begin{aligned} & \Pr_{\mathbf{x}_1, \dots, \mathbf{x}_N \sim D_{\mathcal{L}, s_{1/3}}}[\exists i. \|\mathbf{x}_i\| = \lambda_1(\mathcal{L})] \\ & \geq 1 - \prod_{i=1}^N \Pr_{\mathbf{x}_i \sim D_{\mathcal{L}, s_{1/3}}}[\|\mathbf{x}_i\| \neq \lambda_1(\mathcal{L})] \\ & \geq 1 - (1 - p)^N. \end{aligned}$$

Therefore, it suffices to take $N = \Omega(1/p)$ for the algorithm to succeed with constant probability. $\square$

Corollary 7 (SVP for almost all lattices). *There is a randomized algorithm that for every $n \geq 70$, solves SVP in time $2^{0.635n + o(n)}$ with probability at least $1/2$ on a fraction at least $1 - 0.99^{n/2}$ of random lattices $\mathcal{L}$ according to μ_n .*

Proof. Apply Theorem 11 with $\alpha = 1/0.99$. Note that when $n \geq 70$, we indeed have $\alpha > 2^{1/n}$. Then the fraction of lattices on which the algorithm works is at least $1 - 2^{1-n/2}(1 + o(1)) - \frac{2\alpha^n A(n)}{(\alpha^n - 2)^2} = 1 - 0.99^{n+o(n)}$. Finally, substituting into the runtime complexity gives the result. $\square$

Corollary 8 (SVP for most lattices). *There is a randomized algorithm that for every $k \in \mathbb{N}$ and $n \geq 1$, solves SVP in time $2^{(\frac{1}{2} + \frac{1}{4e \ln 2})n + o(n)} = 2^{0.63269n + o(n)}$ with probability at least $1/2$ on a fraction at least $1 - 2/n^k(1 + o(1))$ of random lattices $\mathcal{L}$ according to μ_n .*

Remark 4. Note that $2^{(\frac{1}{2} + \frac{1}{4e \ln 2})n} = \lim_{\gamma \rightarrow 1} (\frac{2}{\gamma^2} e^{\gamma^2/2e})^{n/2}$, therefore this corollary matches the limit behaviour of Corollary 6 when the approximation factor γ tends to 1.

Proof. Apply Theorem 11 with $\alpha = (n^k + 2 + \sqrt{n^{2k} + 4n^k})^{1/n}$ and note that $\alpha > 2^{1/n}$ by definition. Check that $(\alpha^n - 2)^2 = 2n^k \alpha^n$. Then the fraction of lattices on which the algorithm works is at least $1 - 2^{1-n/2}(1 + o(1)) - \frac{2\alpha^n A(n)}{(\alpha^n - 2)^2} = 1 - 2n^{-k}(1 + o(1))$. The algorithm runs in time $2^{\tau n + o(n)}$ where $\tau = \frac{1}{2} + \frac{\alpha^2}{4e \ln 2}$. Finally check that

$$\begin{aligned} \alpha^2 &= \exp\left(\frac{2}{n} \log(2n^k + 4 + O(n^{-k}))\right) \\ &= \exp\left(\frac{2k \log(n) + 2 \log(2)}{n} + O(n^{-1-k})\right) \\ &= 1 + \frac{2k \log(n) + 2 \log(2)}{n} + O\left(\frac{\log^2 n}{n^2}\right) \\ &= 1 + o(1). \end{aligned}$$

Therefore $\tau = \frac{1}{2} + \frac{1+o(1)}{4e \ln 2} = 0.63269 + o(1)$ which shows the result. $\square$

6.4 GapSVP

Finally, we observe that the strong bounds on λ_1 for a random lattices give a polynomial time algorithm for $(1 + \varepsilon)$ -GapSVP. Although this is more of a folklore result, our theorem gives explicit constants and bounds not found in the literature.

Theorem 12 (GapSVP). *There is a deterministic algorithm that for any $n \geq 1$ and $\gamma = \gamma(n) > 2^{2/n}$, and on a fraction at least $1 - \varepsilon - \varepsilon'$ of random lattices according to μ_n , solves γ -GapSVP in polynomial time (independent of γ), where $\varepsilon = \frac{2\alpha^n A(n)}{(\alpha^n - 2)^2}$, $\varepsilon' = \frac{2\beta^n A(n)}{(2 - \beta^n)^2}$, $\alpha = \sqrt{\gamma}$ and $\beta = 1/\sqrt{\gamma}$ and $A(n)$ is defined in Theorem 8.*

Proof. The algorithm is trivial: on input $\mathcal{L}$ and $r > 0$, accept if $\text{vol}(B_n)^{-1/n} < r\sqrt{\gamma}$ and reject otherwise.

We now analyze the algorithm: let $\alpha = \sqrt{\gamma}$ and note that $\alpha > 2^{1/n}$. Apply Theorem 8 to get that for a fraction at least $1 - \varepsilon$ of lattices $\mathcal{L}$, where $\varepsilon = \frac{2\alpha^n A(n)}{(\alpha^n - 2)^2}$, we have $\lambda_1(\mathcal{L}) \leq \alpha \text{vol}(B_n)^{-1/n}$. Similarly, let $\beta = 1/\sqrt{\gamma}$ and note that $\beta < 2^{-1/n} < 2^{1/n}$. Therefore, for a fraction at least $1 - \varepsilon'$ of lattices $\mathcal{L}$, where $\varepsilon' = \frac{2\beta^n A(n)}{(2 - \beta^n)^2}$, we have $\lambda_1(\mathcal{L}) > \beta \text{vol}(B_n)^{-1/n}$. By a union bound, for a fraction at least $1 - \varepsilon - \varepsilon'$ of lattices $\mathcal{L}$, we have

$$\beta \text{vol}(B_n)^{-1/n} < \lambda_1(\mathcal{L}) \leq \alpha \text{vol}(B_n)^{-1/n}. \quad (12)$$

Assume that the input lattice satisfies (12) and let $r > 0$. There are two cases to consider:

- If $\lambda_1(\mathcal{L}) \leq r$ then $\text{vol}(B_n)^{-1/n} < r/\beta = r\sqrt{\gamma}$ so the algorithm accepts.
- If $\lambda_1(\mathcal{L}) \geq \gamma r$ then $\text{vol}(B_n)^{-1/n} \geq r\gamma/\alpha = r\sqrt{\gamma}$ so the algorithm rejects.

Therefore the algorithm solves γ -GapSVP on $\mathcal{L}$. The running time is clearly polynomial. $\square$

Since the statement of Theorem 12 is quite hard to decipher, we give a weaker result in the case of a constant approximation factor, which can be arbitrarily close to 1.

Corollary 9 (GapSVP on almost all lattices). *There is a deterministic algorithm that for any $\gamma > 1$ solves γ -GapSVP for $n \geq \frac{4 \ln 2}{\ln \gamma}$ in polynomial time (independent of γ) and on a fraction at least $1 - 10\gamma^{-n/2}(1 + o(1))$ of random lattices according to μ_n .*

Proof. Apply Theorem 12 with γ . Note that since $n \geq 4 \ln(2)/\ln(\gamma)$, we have $\gamma \geq 4^{2/n} > 2^{2/n}$. Therefore the algorithm solves γ -GapSVP on a fraction at least $1 - \varepsilon - \varepsilon'$ of lattices. Now observe that $\alpha^n = \gamma^{n/2} > 4$, therefore^{††}

$$\varepsilon = \frac{2\alpha^n A(n)}{(\alpha^n - 2)^2} \leq \frac{8A(n)}{\alpha^n} = 8\gamma^{-n/2}(1 + o(1)).$$

The bound of ε' is easier since $\beta = 1/\sqrt{\gamma} \leq 1$ so

$$\varepsilon' = \frac{2\beta^n A(n)}{(2 - \beta^n)^2} \leq 2\beta^n A(n) = 2\gamma^{-n/2}(1 + o(1)).$$

$\square$

Acknowledgments.

This work was supported by the French Agence Nationale de la Recherche through the France 2030 program under grant agreement No. ANR-22-PETQ-0008 PQ-TLS. We thank the anonymous reviewers of CRYPTO 2025 for finding an error in the previous version of the paper.

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^{††}We use that $\frac{x}{(x-2)^2} \leq \frac{4}{x}$ for $x \geq 4$ which follows by a simple analysis.

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